

Coupling Interpretable Allocation, Model-Predictive Control, and Tail-Aware Forecasting for Live Portfolio Management

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Outline

Live portfolio management couples three problems:
allocation, the timing of rebalancing, and evaluation beyond the historical record.

1	**Allocation**	What to hold. Maximize utility of a score γ for each stock, within trading limits.
2	**Rebalancing**	When to trade. Re-plan at the 21-day horizon, a band exit, or a drawdown stop; real-data backtest, 2025–26.
3	**Market generator**	Evaluation. A backtest is one historical path; JumpHMM generates typical and stressed futures.

Maximize utility subject to trading limits (convex cone)

Each rebalance chooses n_i shares of each name i to solve:

$$\max_{n \geq 0} \; U(n;\gamma)$$ — utility U (γ_i: score of name i, next slide)

$$\text{s.t.} \; \sum_i n_i S_i \leq W$$ — budget W (S_i: price of name i)

$$w_i \leq \bar{w}$$ — single-name cap $\bar{w}$ (w_i: weight of name i)

$$\sqrt{w^\top \Sigma_g \, w} \leq \sigma_{\text{SPY}}$$ — volatility cap σ_{SPY} (Σ_g: growth-rate covariance)

$$\bar{c} \, \| n - n^{\text{prev}} \|_1 \leq B$$ — turnover budget B ($\bar{c}$: half spread per share)

$$\beta^\top w \leq \bar{\beta}$$ — β cap $\bar{\beta}$, hedge on (β: beta to SPY)

	Cobb–Douglas	CES on shares	CES on weights	Stone–Geary
$U(n;\gamma)$, over $\gamma_i > 0$	$\sum \gamma_i \ln n_i$	$(\sum \gamma_i n_i^\rho)^{1/\rho}$	$(\sum \gamma_i w_i^\rho)^{1/\rho}$	$\sum \gamma_i \ln(n_i - b_i)$
budget alone	$w_i \propto \gamma_i$	$n_i \propto (\gamma_i / S_i)^\eta$	$w_i \propto \gamma_i^\eta$	$(n_i - b_i)\, S_i \propto \gamma_i$

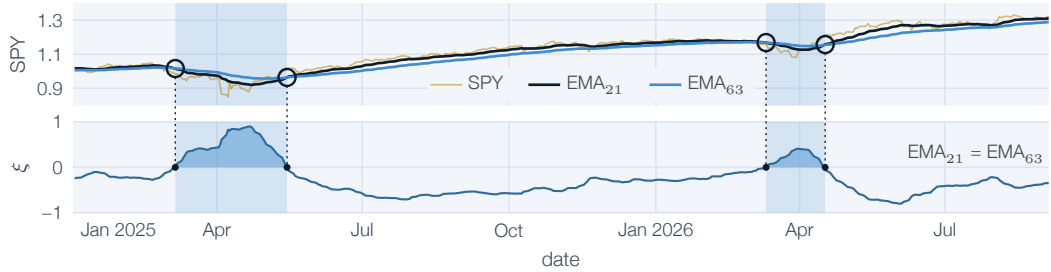
$\rho = 1 - 1/\eta$, $\eta = 0.5$; $b_i S_i = 1\%$ of W. Benchmarks under the same limits: minimum variance, tangency, 130/30.

The score γ: expected growth, tilted by the market

$$\gamma_i = \tanh\left(\frac{\alpha_i + \beta_i\,\tilde{g}_M}{\max(|\beta_i|,\,0.1)^{\xi}}\right)$$

expected growth (α_i, β_i: fit of name i to SPY; $\tilde{g}_M$: SPY's growth rate, 252-day EMA)

β penalty (ξ: market state, >0 when $\text{EMA}_{21} < \text{EMA}_{63}$)

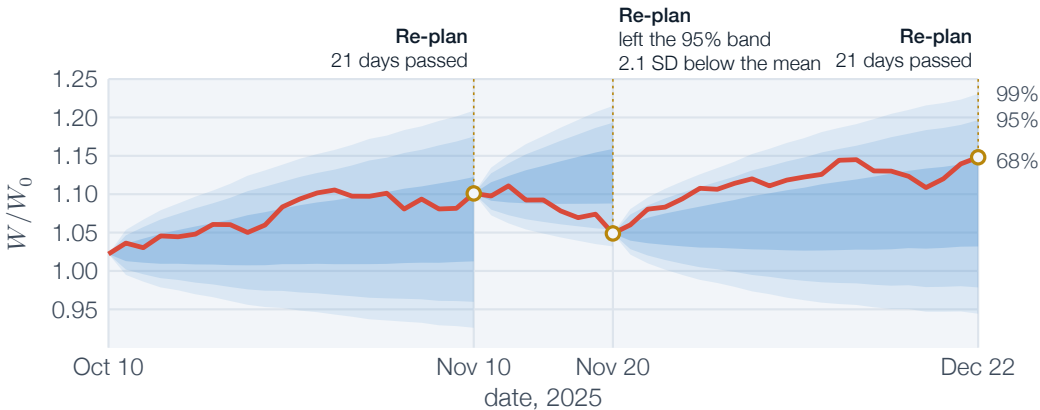


Weighting	γ enters $U(n;\gamma)$: names with $\gamma_i > 0$ are held long, and 130/30 shorts $\gamma_i < 0$.
Selection	the 13 highest γ overall, or the 3 highest in each sector.

$\xi = -G\,(\text{EMA}_{21}/\text{EMA}_{63} - 1)$ from SPY's 21- and 63-day exponential averages; $|\xi| \le 1$ on 99% of 2014–2024 days.

Rebalancing triggers: horizon, forecast band, drawdown

Forecast: multi-asset GBM, 1,000 paths, 21 trading days ahead. Plant: real prices.



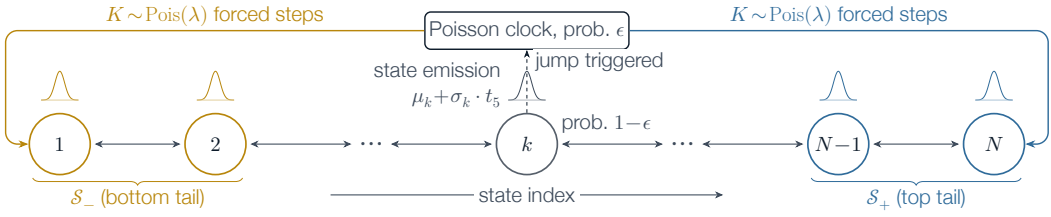
Hand-picked 13, Cobb–Douglas, hedge off; over Jan 2025 – Sep 2026, 27 re-plans, 10 before the horizon.

Stock selection matters more than the weighting rule

Real-data backtest, Jan 2025 – Sep 2026, 373 names. Each row is a selection rule.



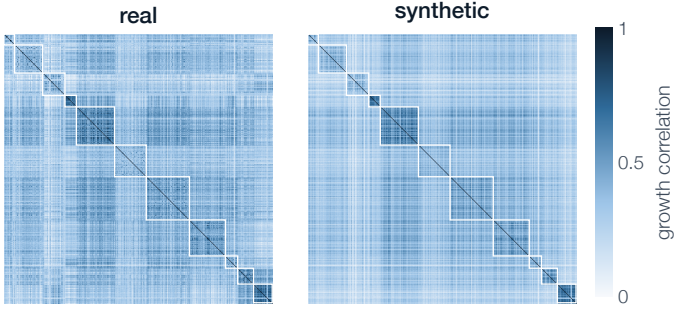
JumpHMM: a market generator with fat tails



The controller forecasts with GBM; JumpHMM generates the test markets. Alswaidan and Varner, arXiv:2603.10202.

Synthetic names move together like the real ones

$$g_i(t) = \underbrace{\alpha_i}_{\text{trend}} + \underbrace{\beta_i(t)\, g_m(t)}_{\text{market: 1 JumpHMM}} + \underbrace{\delta_i\, f_s(t)}_{\text{sector: 11 JumpHMMs}} + \underbrace{s_i\, \tilde{\varepsilon}_i(t)}_{\text{own: 373 JumpHMMs}}$$



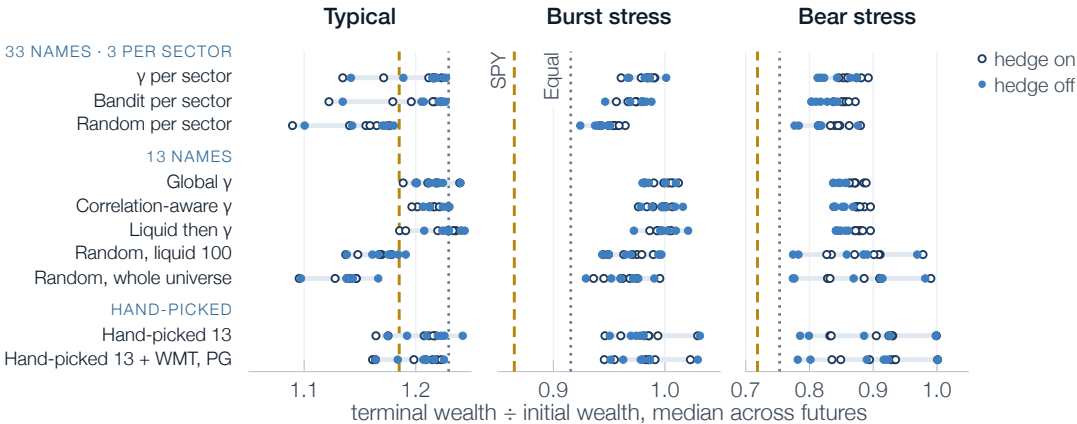
mean pairwise correlation

	within a sector	across sectors
real	0.47	0.35
synthetic	0.45	0.31

Real: 373 names, 2014–2024. Synthetic: one simulated path, same dates. Names sorted by sector; sectors outlined.

Simulated sell-offs: every median ends above SPY

600 typical JumpHMM futures of Jan 2025 – Sep 2026, and 200 in each stress set.



Stress sets: of 10,000 paths, the 200 deepest SPY drawdowns with a jump burst, and the 200 without one.

Key takeaways

- **Allocation: stock selection mattered more than the weighting rule.**
  With the hedge off, seven weighting rules on one 33-name basket ended at 1.09–1.23 times initial wealth; different baskets ended at 1.03–1.70.

- **Rebalancing: re-plans came at the horizon, a band exit, or a drawdown stop.**
  About 27 re-plans in 419 trading days, a third of them before the 21-day horizon.

- **Evaluation: in both sell-off sets, every median ended above SPY.**
  JumpHMM reproduces fat tails, but its volatility clustering fades after a day. With the hedge off, 13 names picked by γ beat 13 random names in 62–71% of typical futures; in bear futures, random picks did as well or better for most rules.

A paper-trading version of this system is running now; its results are not part of this talk.

Acknowledgements

Abdulrahman Alswaidan, MS 2026
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helpful discussions

Evan Vu, class of 2026, now at Bridgewater
an early version of the Cobb–Douglas allocation and the bandit picker

We also thank Alpaca Markets.

Paper

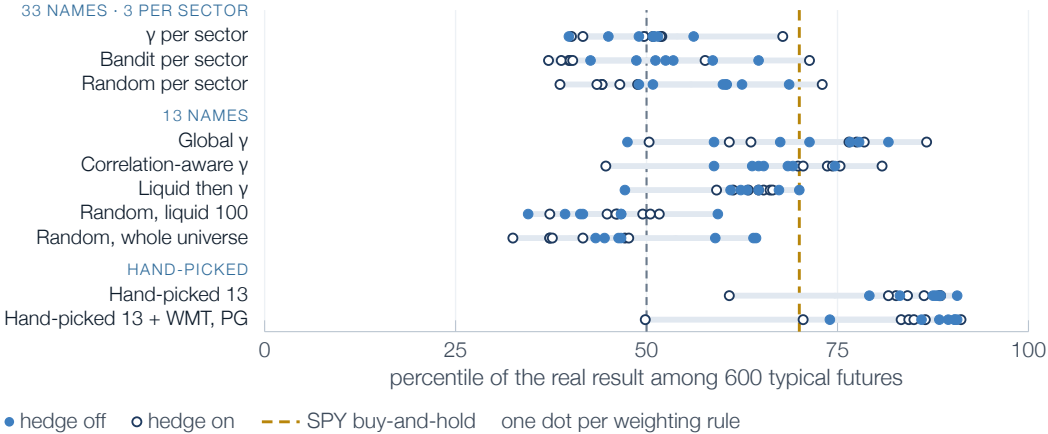
JumpHMM paper
arXiv:2603.10202

Code

github.com/varnerlab/INFORMS-Annual-Meeting-2026-Varnerlab

Real results as percentiles of typical futures

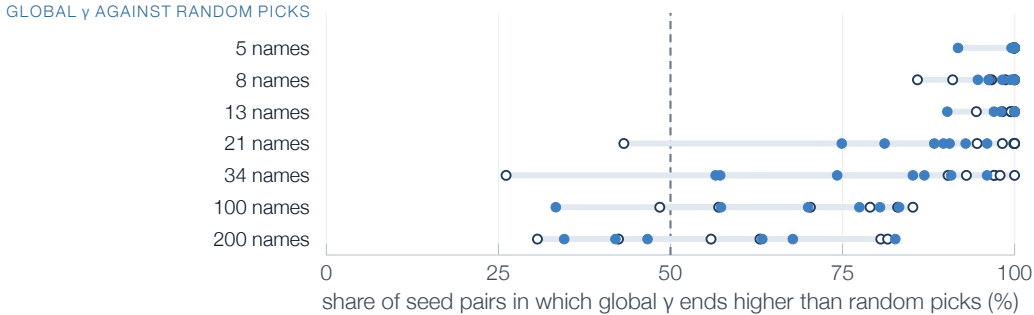
Each real result among 600 typical JumpHMM futures. Right of 50 beats the median future.



Real results use the simulated runs' seeds: 5660–5664 for random and bandit pickers, 5660 for the rest.

Global γ against random picks, 5 to 200 names

The K highest γ in the universe against K random names, from 5 to 200 names. At 50%, the two picks end higher equally often.



● hedge off ○ hedge on one dot per weighting rule

Real-data backtest, 30 runs each. Stone–Geary left out at 100 and 200 names: its 1%-per-name floor uses the whole budget at 100 names and twice the budget at 200.

Terminal wealth by picker and utility: hedge off

Picker	CD	CES	CESw	SG	MinVar	Tangent	LS
γ per sector	1.230	1.219	1.221	1.234	1.086	1.227	1.110
Bandit per sector	1.226	1.234	1.223	1.239	1.162	1.229	1.140
Random per sector	1.205	1.183	1.201	1.183	1.128	1.158	1.127
Global γ	1.520	1.384	1.460	1.441	1.355	1.341	1.248
Correlation-aware γ	1.352	1.386	1.421	1.380	1.284	1.404	1.219
Liquid then γ	1.333	1.310	1.379	1.350	1.238	1.361	1.305
Random, liquid 100	1.216	1.173	1.211	1.212	1.158	1.274	1.152
Random, whole universe	1.098	1.132	1.138	1.112	1.104	1.160	1.031
Hand-picked 13	1.548	1.571	1.614	1.631	1.443	1.697	1.549
Hand-picked 13 + WMT, PG	1.616	1.646	1.637	1.639	1.362	1.642	1.622
SPY buy-and-hold	1.317	1.317	1.317	1.317	1.317	1.317	1.317
equal-weight universe	1.296	1.296	1.296	1.296	1.296	1.296	1.296

Terminal wealth ÷ initial wealth, median of 30 runs. Real-data backtest, Jan 2025 – Sep 2026.
CD Cobb–Douglas · CES on shares · CESw CES on weights · SG Stone–Geary · MinVar minimum variance · Tangent the tangency portfolio · LS 130/30 long-short Cobb–Douglas.

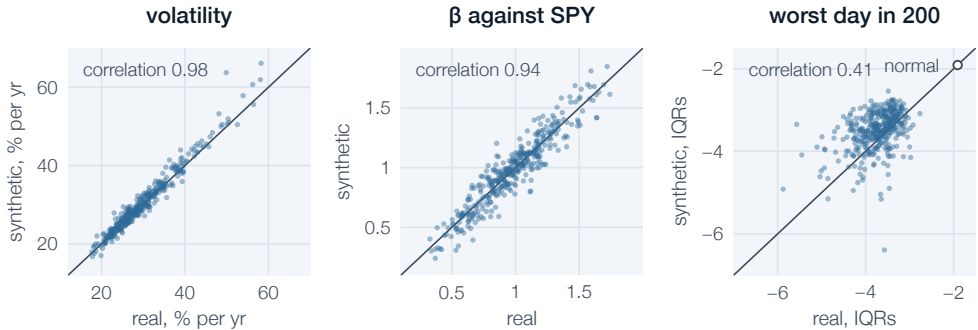
Terminal wealth by picker and utility: hedge on

Picker	CD	CES	CESw	SG	MinVar	Tangent	LS
γ per sector	1.212	1.232	1.223	1.218	1.073	1.285	1.130
Bandit per sector	1.125	1.140	1.167	1.194	1.166	1.214	1.120
Random per sector	1.108	1.087	1.114	1.155	1.155	1.092	1.092
Global γ	1.503	1.305	1.464	1.461	1.203	1.589	1.283
Correlation-aware γ	1.409	1.429	1.479	1.447	1.213	1.366	1.292
Liquid then γ	1.322	1.285	1.347	1.331	1.184	1.317	1.336
Random, liquid 100	1.167	1.156	1.174	1.178	1.123	1.155	1.112
Random, whole universe	1.069	1.106	1.093	1.115	1.079	1.062	0.994
Hand-picked 13	1.554	1.517	1.517	1.511	1.233	1.582	1.586
Hand-picked 13 + WMT, PG	1.564	1.571	1.543	1.542	1.159	1.383	1.630
SPY buy-and-hold	1.317	1.317	1.317	1.317	1.317	1.317	1.317
equal-weight universe	1.296	1.296	1.296	1.296	1.296	1.296	1.296

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Each synthetic name resembles its real counterpart

$$g_i(t) = \underbrace{\alpha_i}_{\text{trend}} + \underbrace{\beta_i(t)\, g_m(t)}_{\text{market: 1 JumpHMM}} + \underbrace{\delta_i\, f_s(t)}_{\text{sector: 11 JumpHMMs}} + \underbrace{s_i\, \tilde{\varepsilon}_i(t)}_{\text{own: 373 JumpHMMs}}$$



Each dot is a name, real 2014–2024 against one simulated path. IQR: interquartile range of daily growth.